



# DATABASE MANAGEMENT SYSTEMS (IT 2040)

## LECTURE 03- SCHEMA REFINEMENT



# LECTURE CONTENT

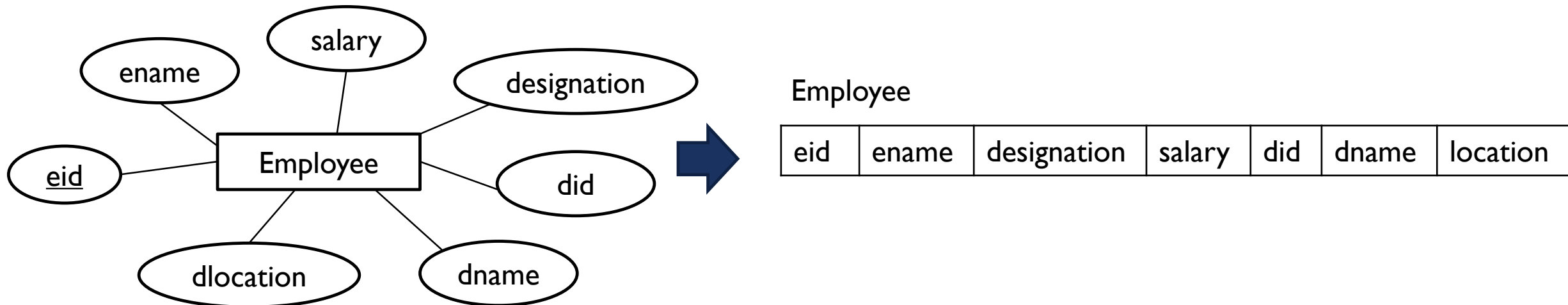
- Why schema refinement?
- Properties of good decomposition
- Functional dependencies
- Computing keys of relations
- Normalization and normal forms

# LEARNING OUTCOMES

- Explain the pitfalls of incorrect grouping of attributes in relations.
- Explain the properties of a good decomposition.
- Compute keys from a given set of FDs in a relation
- Normalize a given schema to BCNF.

# WHY SCHEMA REFINEMENT?

- The relations resulted through the logical database design may not be very good if your conceptual database design is not good.
- For example, what is wrong with the following schema resulted through mapping of an ER diagram



# WHY SCHEMA REFINEMENT? (CONTD.)

- Schemas such as in the previous slide lead to several anomalies while inserting, updating and deleting and wasting of space due to redundancies of data.
- Can you spot where data are duplicated and issues may cause during insert/update & delete?

| <u>eid</u> | Ename    | Designation | Salary | did | dname       | location |
|------------|----------|-------------|--------|-----|-------------|----------|
| 1000       | Ajith    | Lecturer    | 60000  | 1   | Academic    | malabe   |
| 1001       | Sunil    | Executive   | 45000  | 3   | Maintenance | Kandy    |
| 1002       | Kamal    | Lecturer    | 75000  | 1   | Academic    | malabe   |
| 1003       | Piyumi   | Manager     | 50000  | 2   | Admin       | metro    |
| 1004       | Roshan   | Lecturer    | 35000  | 1   | Academic    | malabe   |
| 1005       | Nuwan    | Lecturer    | 80000  | 1   | Academic    | malabe   |
| 1006       | Jayamini | Assistant   | 25000  | 2   | Admin       | metro    |
| 1007       | Nishani  | Lecturer    | 42000  | 1   | Academic    | malabe   |
| 1008       | Amal     | Assistant   | 28000  | 4   | ITSD        | Matara   |

# WHY SCHEMA REFINEMENT? (CONTD.)

## ■ Insertion Anomaly

- Inserting a new employee to the emp table
  - Department information is repeated (ensure that correct department information is inserted).
- Inserting a department with no employees
  - Impossible since eid cannot be null

## ■ Deletion Anomaly

- Deleting the last employee from the department will lead to losing information about the department

## ■ Update Anomaly

- Updating the department's location needs to be done for all employees working for that department

# WHY SCHEMA REFINEMENT? (CONTD.)

- To solve the issues discussed previously, we should decompose the relations to smaller relations.
- For example, we can decompose the emp relation discussed previously into two relations as below to overcome the issues of redundancies and anomalies.

| <u>eid</u> | Ename    | Designation | Salary | did |
|------------|----------|-------------|--------|-----|
| 1000       | Ajith    | Lecturer    | 60000  | 1   |
| 1001       | Sunil    | Executive   | 45000  | 3   |
| 1002       | Kamal    | Lecturer    | 75000  | 1   |
| 1003       | Piyumi   | Manager     | 50000  | 2   |
| 1004       | Roshan   | Lecturer    | 35000  | 1   |
| 1005       | Nuwan    | Lecturer    | 80000  | 1   |
| 1006       | Jayamini | Assistant   | 25000  | 2   |
| 1007       | Nishani  | Lecturer    | 42000  | 1   |
| 1008       | Amal     | Assistant   | 28000  | 4   |

| did | dname       | location |
|-----|-------------|----------|
| 1   | Academic    | malabe   |
| 2   | Admin       | metro    |
| 3   | Maintenance | Kandy    |
| 4   | ITSD        | Matara   |

## WHY SCHEMA REFINEMENT? (CONTD.)

- Random decompositions however, may introduce new problems.
- Two properties that could be looked up to ensure that the relations resulted from decomposition are good are as follows:
  - Loss-less join property
  - Dependency preserving property

# WHY SCHEMA REFINEMENT? (CONTD.)

- **Loss-less join property** : the property enables recovery of original relation from a set of smaller relations resulted through decomposition.
  - For example, suppose S is decomposed into R<sub>1</sub> and R<sub>2</sub>. When we join R<sub>1</sub> and R<sub>2</sub> do we get S? if so, we say the decomposition is loss-less.

S

| S  | P  | D  |
|----|----|----|
| S1 | P1 | D1 |
| S2 | P2 | D2 |
| S3 | P1 | D3 |

R<sub>1</sub>

| S  | P  |
|----|----|
| S1 | P1 |
| S2 | P2 |
| S3 | P1 |

R<sub>2</sub>

| P  | D  |
|----|----|
| P1 | D1 |
| P2 | D2 |
| P1 | D3 |

- **Dependency preserving property** : enables to enforce any constraint on the original relation simply enforcing some constraints on each of the smaller relation.

# SCHEMA REFINEMENT

- Schema refinement can be considered a systematic process for analyzing a relational schema with the aim of minimizing redundancies and minimizing insertion, deletion and update anomalies.
- The process performs a series of tests called **normal form tests** to check whether the schemas meet certain conditions, and those relations that are unsatisfactory are decomposed in to smaller relation.
- Normalization is based on functional dependencies

# FUNCTIONAL DEPENDENCY IN GENERAL TERMS

- Functional dependency is a relationship that exists when one attribute uniquely determine another attribute.
- For example: Suppose we have a student table with attributes: id name, age.
  - Here id attribute uniquely identifies the name attribute of student table because if we know the student id we can tell the student name associated with it.
  - This is known as functional dependency and can be written as  $id \rightarrow name$  or in words we can say name is functionally dependent on id.
- Redundancies in relations are based on functional dependencies

# FUNCTIONAL DEPENDENCIES

- Mathematical definition of functional dependency (FD) :
  - A functional dependency, denoted by  $X \rightarrow Y$ , where  $X$  and  $Y$  are sets of attributes in relation  $R$ , specifies the following constraint:
    - Let  $t_1$  and  $t_2$  be tuples of relation  $R$  for any given instance  
Whenever  $t_1[X] = t_2[X]$  then  $t_1[Y] = t_2[Y]$   
where  $t_i[X]$  represents the values for  $X$  in tuple  $t_i$

# ACTIVITY

- Consider a person entity. With respect to attributes you know a person holds write three functional dependencies that can exist in a person table.
- Exchange what you have written with your peers.
- What have they written?

# KEYS AND FDS

- A key constraint is a special case of a FD where the attributes in the key play the role of  $X$  and the set of all attributes play the role of  $Y$ .
- Normalization process analyzes schemas based on keys and on the functional dependencies among their attributes.
- Thus, to start normalization it is essential to find keys of a given relation.
- Attribute closure of an attribute set  $X$ , denoted by  $X^+$  can be defined as set of all attributes which can be functionally determined from it.
- If  $X^+ = \text{all attributes}$ , then  $X$  is a key.

# COMPUTING KEYS USING ARMSTRONG AXIOMS

- Attribute closure could be computed using Armstrong Axioms.
- $X, Y, Z$  are sets of attributes:

Reflexivity: If  $X \subseteq Y$ , then  $Y \rightarrow X$

Augmentation: If  $X \rightarrow Y$ , then  $XZ \rightarrow YZ$  for any  $Z$

Transitivity: If  $X \rightarrow Y$  and  $Y \rightarrow Z$ , then  $X \rightarrow Z$

- Couple of additional rules (that follow from Armstrong Axioms):

Union: If  $X \rightarrow Y$  and  $X \rightarrow Z$ , then  $X \rightarrow YZ$

Decomposition: If  $X \rightarrow YZ$ , then  $X \rightarrow Y$  and  $X \rightarrow Z$

## COMPUTING KEYS USING AMSTRONG AXIOMS (CONTD.)

- Consider a relation  $R(A, B, C, D)$ , with the following set of functional dependencies over  $R$ :
    - $F = \{A \rightarrow B, B \rightarrow C, B \rightarrow D\}$ 
      - $A \rightarrow A$  (reflexivity rule)
      - $A \rightarrow B$  (given)
      - $A \rightarrow B$  and  $B \rightarrow C$  then  $A \rightarrow C$  (transitivity)
      - $A \rightarrow B$  and  $B \rightarrow D$  then  $A \rightarrow D$  (transitivity)
- $A \rightarrow ABCD$  (Union rule) ,  $[A]^+ = \{ABCD\}$
- $B^+ = \{BCD\}$  ,  $C^+ = \{C\}$  ,  $D^+ = \{D\}$
- Therefore  $A$  is the key

# ACTIVITY

- Consider a relation  $R(A, B, C, D, E)$ , with the following set of functional dependencies over  $R$ :
- $\{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$
- Compute the keys for relation  $R$ .

# REVISIT TO SOME DEFINITIONS

- **Superkey**: Set of attributes  $S$  in relation  $R$  such that no two distinct tuples  $t_1$  and  $t_2$  will have  $t_1[S] = t_2[S]$
- **Key**: A key is a superkey with the additional property that removal of any attributes from the key will not satisfy the key condition
- **Candidate Key**: Each key of a relation is called a candidate key
- **Primary Key**: A candidate key is chosen to be the primary key
- **Prime Attribute**: an attribute which is a member of a candidate key
- **Nonprime Attribute**: An attribute which is not prime

# NORMAL FORMS

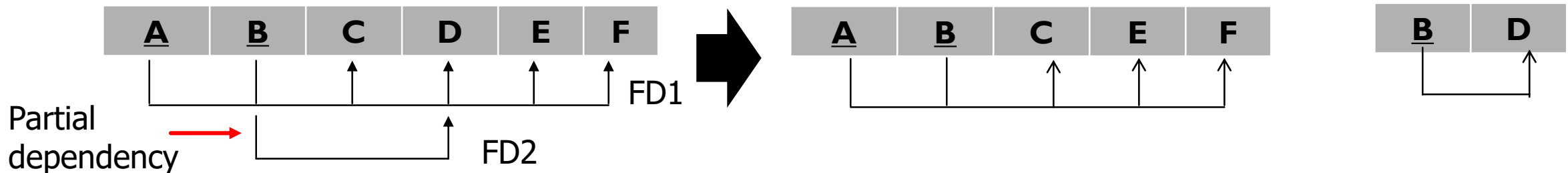
- Normal forms refers to a series of tests performed on relational schemas to improve their goodness.
- We discuss four normal forms namely,
  - 1<sup>st</sup> Normal Form
  - 2<sup>nd</sup> Normal Form
  - 3<sup>rd</sup> Normal Form
  - Boyce Codd Normal Form
- Test for each normal form is performed in a top-down fashion.
- The Normal form of a relation refers to the highest normal form condition it meets.

# 1<sup>ST</sup> NORMAL FORM

- A relation R is in first normal form (1NF) if domains of all attributes in the relation are *atomic* (simple & indivisible).
- 1NF is now considered to be part of the formal definition of a relation in the relational model since it allows only atomic values and disallows multivalued attributes and composite attributes.

# 2<sup>ND</sup> NORMAL FORM

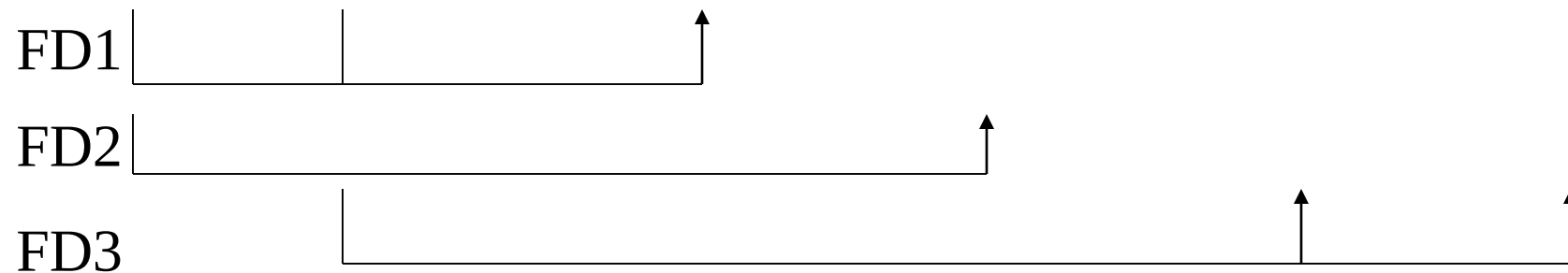
- A relation R is in second normal form (2NF) if every nonprime attribute A in R is not partially dependent on any key of R.
- Second normal form (2NF) is based on the concept of full functional dependency.
  - A functional dependency  $X \rightarrow Y$  is a full functional dependency if removal of any attribute A from X means that the dependency does not hold any more.
  - For example, in a relation R (ABCDE) where  $AB \rightarrow \{ABCDE\}$ , if  $A \rightarrow C$ ,  $A \rightarrow C$  is a partial dependency (not fully functional dependent)
- To normalize the relation to 2NF decomposition is performed as follows.



# ACTIVITY

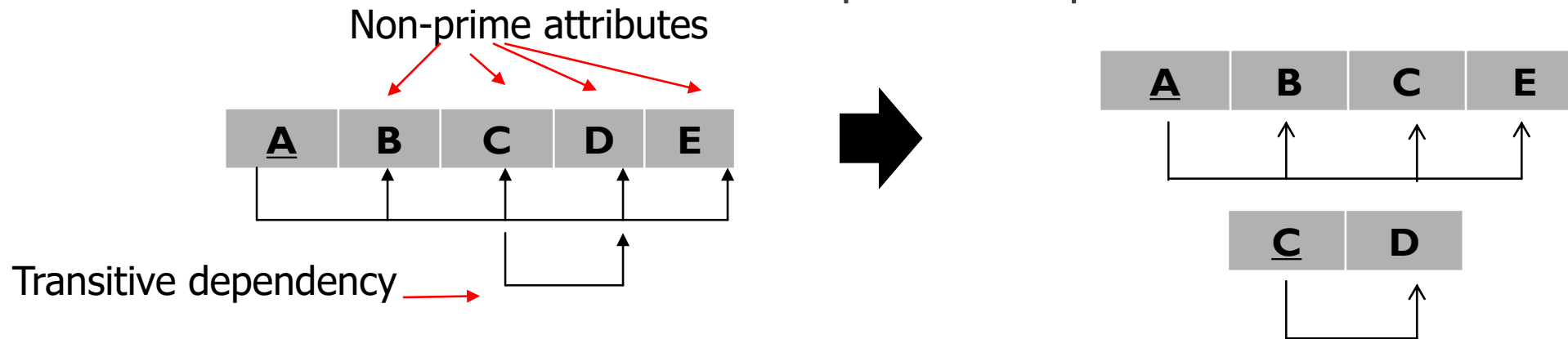
- What normal form the relation in the slide is? If the relation is not in 2NF normalize the relation to 2NF.

|          |            |             |       |       |       |     |
|----------|------------|-------------|-------|-------|-------|-----|
| EMP_PROJ | <u>NIC</u> | <u>PNUM</u> | HOURS | ENAME | PNAME | LOC |
|----------|------------|-------------|-------|-------|-------|-----|



# 3<sup>RD</sup> NORMAL FORM

- A relation R is in 3<sup>rd</sup> normal form (3NF) if every
  - R is in 2NF, and no nonprime attribute is transitively dependent on any key
- Third normal form is based on the concept of transitive dependency.
  - A functional dependency  $X \rightarrow Y$  in a relational schema R is a transitive dependency if there is a set of non-prime attributes Z where both  $X \rightarrow Z$  and  $Z \rightarrow Y$  hold.
- To normalize the relation to 2NF decomposition is performed as follows.

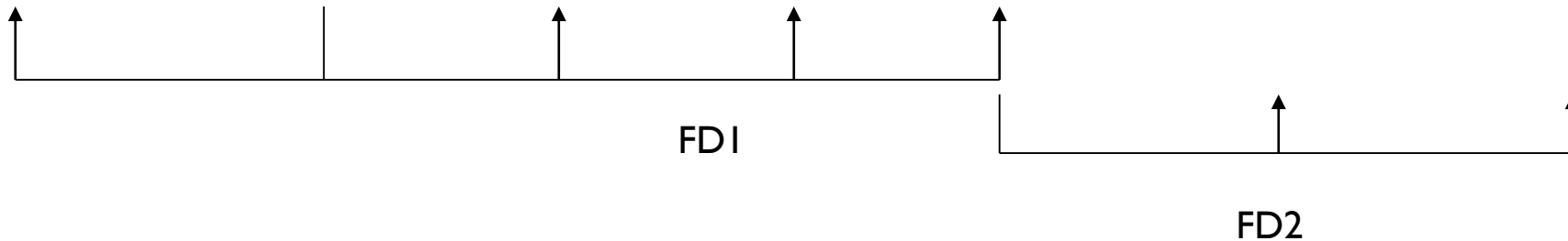


# ACTIVITY

- What normal form the relation in the slide is?
- If the relation is not in 3NF normalize the relation to 3NF.

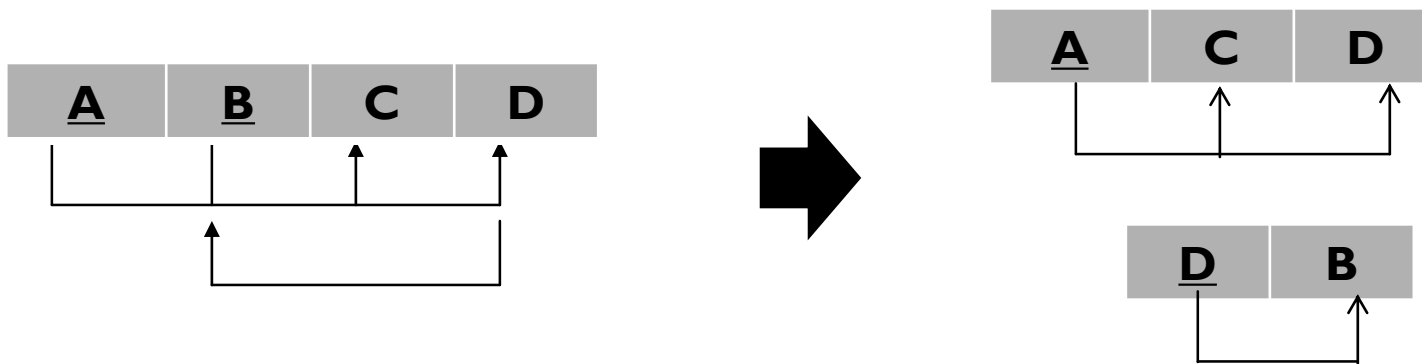
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|       |            |       |     |      |       |      |
|-------|------------|-------|-----|------|-------|------|
| ENAME | <u>SSN</u> | BDATE | ADD | DNUM | DNAME | DMGR |
|-------|------------|-------|-----|------|-------|------|



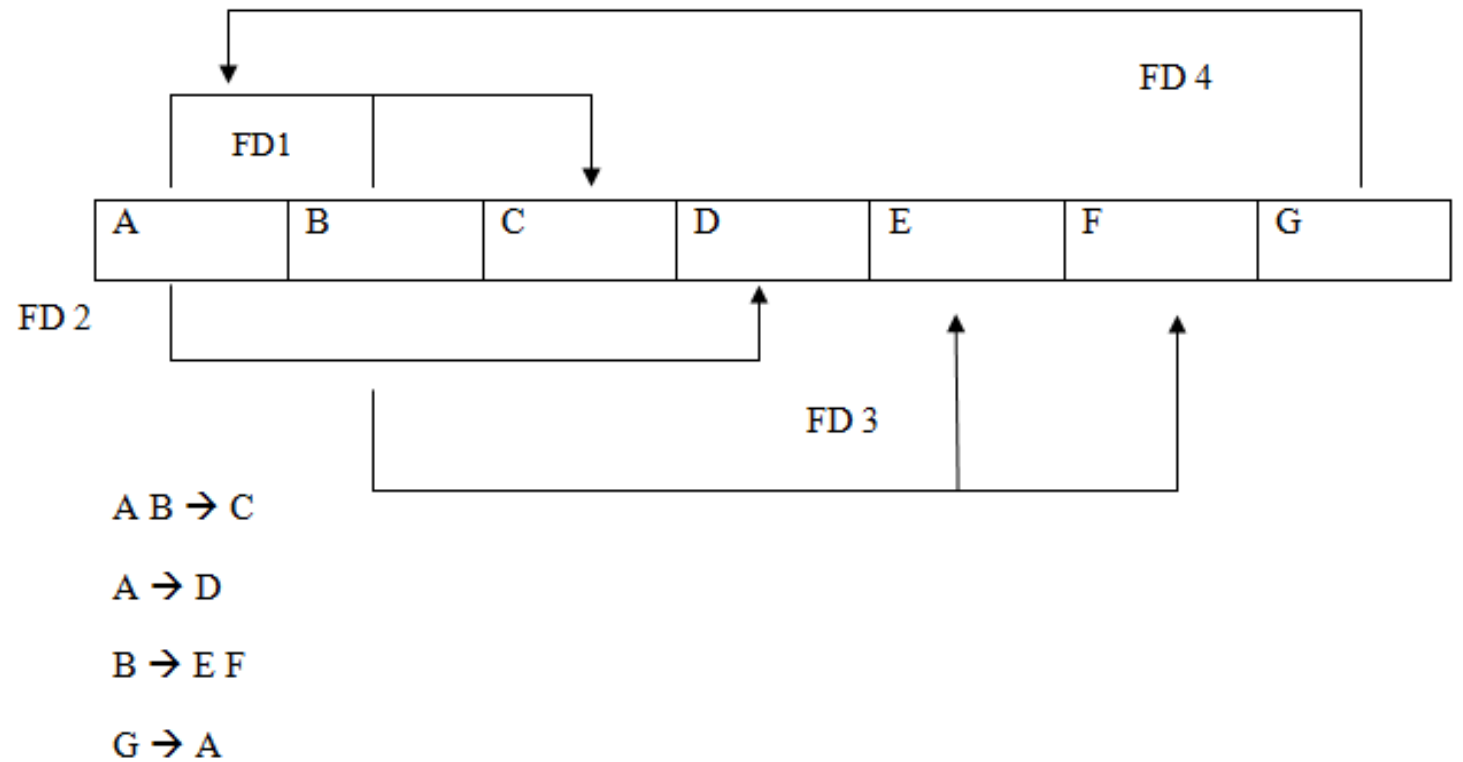
# BOYCE-CODD NORMAL FORM

- A relation schema is in Boyce-Codd Normal Form
  - If every nontrivial functional dependency  $X \rightarrow A$  hold in R, then X is a superkey of R
- Decomposition into BCNF:
  - Consider relation R with FDs F. If  $X \rightarrow Y$  violates BCNF, decompose R

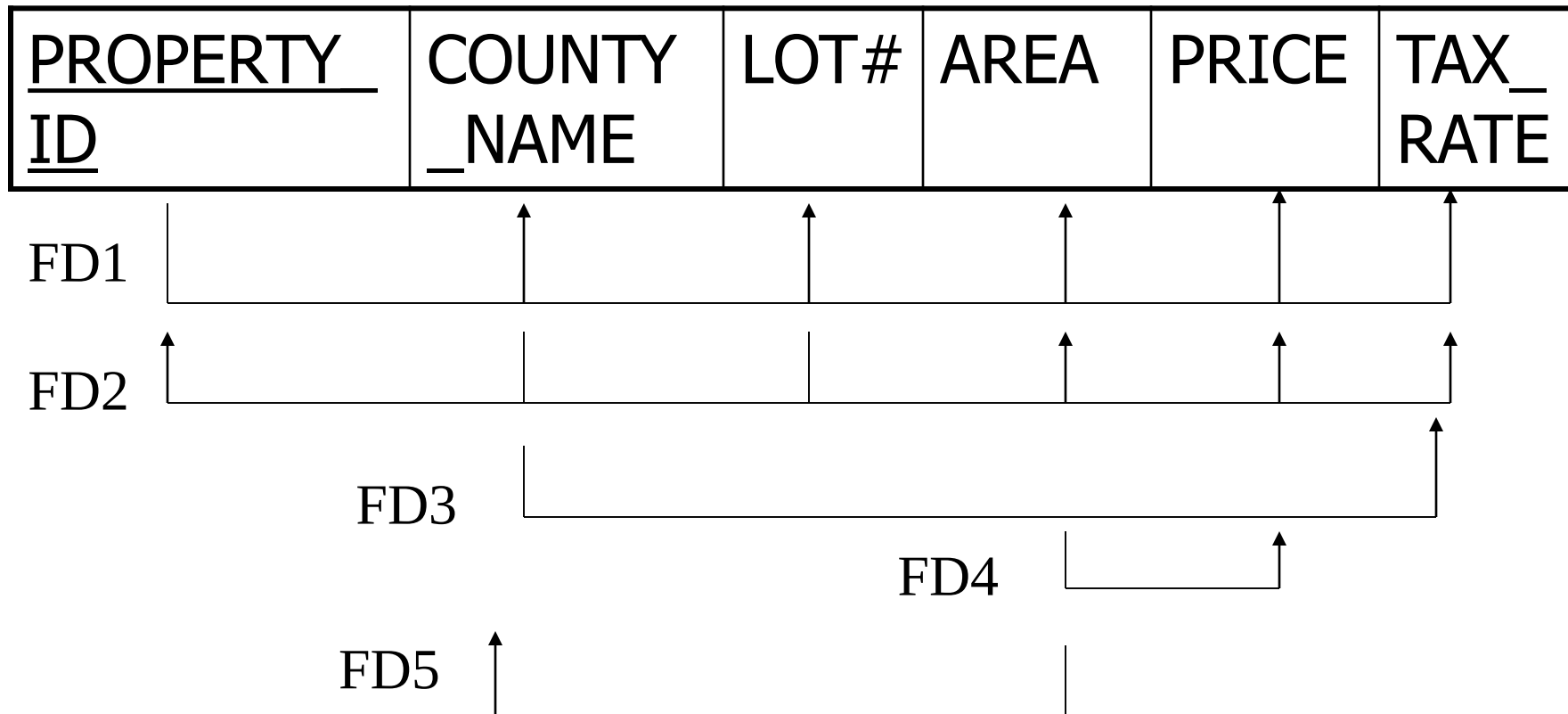


# ACTIVITY

- Consider the following relational schema for R:  
 $R(A, B, C, D, E, F, G)$
- AB is the primary key in the relation.
- What normal form is the relation in?
- If R is not in BCNF, convert it to BCNF.



# ACTIVITY



# WHAT YOU HAVE TO DO BY NEXT WEEK

- Try out the self-test questions on the course web.
- Complete the tutorial.
- Read Chapter